Crop Care

Soil, planting, water, and in-season management for US growers

A detailed reference guide • United States
Compiled October 2026

Note: *This guide compiles general, widely accepted agronomic practice for US growers. Conditions vary by state, soil, and season — confirm specifics with your local Cooperative Extension office (find yours at nifa.usda.gov/land-grant-colleges-and-universities-partner-website-directory) and always follow current product labels and local regulations.*

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1. Soil Health & Preparation

Healthy soil is the foundation of every successful crop. It supplies water, nutrients, and physical anchoring for roots, and a biologically active soil suppresses disease and cycles nutrients efficiently. Good crop care begins months before planting, with assessment and preparation of the seedbed.

Assessing your soil

- **Soil texture.** The proportion of sand, silt, and clay determines water-holding capacity and drainage. Loam (a balanced mix) is ideal for most crops; sandy soils drain fast and need more frequent water and nutrients; clay soils hold water but can waterlog and compact.
- **Soil pH.** Most crops perform best between pH 6.0 and 7.0. Test every 2–3 years. Acidic soils are corrected with agricultural lime; alkaline soils with elemental sulfur or acidifying fertilizers.
- **Organic matter.** Aim for 3–5% in cropland. Organic matter improves structure, water retention, and nutrient supply. Build it with cover crops, manure, compost, and reduced tillage.
- **Compaction.** Check for hardpans with a penetrometer or by digging. Compacted layers restrict roots and water movement; relieve with deep-rooted cover crops or subsoiling when soil is dry.

Seedbed preparation

A good seedbed is firm below and loose on top, free of large clods, and level enough for uniform seed placement. Over-working soil destroys structure and invites crusting and erosion, so prepare only as much as the crop needs. For small seeds (lettuce, carrots, alfalfa) a finer, firmer bed is required than for large seeds (corn, beans).

Note: *Avoid working wet soil. Tilling or driving on soil that is too wet smears and compacts it, creating clods and dense layers that persist for years. Squeeze a handful — if it ribbons and stays in a ball, it is too wet.*

2. Tillage Systems

Tillage prepares the seedbed, incorporates residue and amendments, and controls weeds, but it also burns organic matter and exposes soil to erosion. The trend across US agriculture is toward reduced and no-till systems that conserve soil and moisture.

System	Description	Best for
Conventional / full tillage	Plow plus multiple secondary passes; buries nearly all residue.	Heavy clay, fields needing leveling or heavy residue incorporation.
Reduced / minimum tillage	Fewer passes; leaves 15–30% residue on surface.	A middle path balancing weed control and soil protection.
Strip-till	Tills only narrow strips where seed is placed; between-row soil undisturbed.	Row crops (corn, soy) in cold or erosion-prone soils.
No-till	Seed drilled directly into undisturbed soil and residue.	Erosion control, moisture conservation, building organic matter.

No-till and strip-till reduce fuel, labor, and erosion and build soil carbon, but they rely more heavily on herbicides and careful residue management, and soils warm more slowly in spring. Match the system to your soil, climate, equipment, and crop.

3. Planting Fundamentals

Seed quality & selection

- Buy certified seed with a known germination rate and, for hybrids, documented disease and pest resistance.
- Match variety maturity to your growing-season length (see the companion Sowing Seasons guide) so the crop finishes before frost.
- Select disease- and pest-resistant varieties suited to local pressures — this is the cheapest and most durable form of crop protection.
- Consider treated seed (fungicide/insecticide coatings) where early-season soil pests or damping-off are a recurring problem.

Depth, spacing & population

Plant to the depth appropriate for the seed size and soil moisture — generally 2–3 times the seed's diameter. Too shallow risks drying out; too deep wastes seed energy and delays emergence. Uniform depth gives uniform emergence, which is one of the strongest predictors of yield.

Crop	Typical depth	Row / plant spacing
Corn	1.5–2 in	30 in rows; 30,000–36,000 plants/acre
Soybean	1–1.5 in	15–30 in rows; 120,000–160,000/acre
Wheat	1–1.5 in	6–8 in drill rows
Cotton	0.5–1 in	30–40 in rows
Tomato (transplant)	set to first leaves	18–36 in apart, 3–4 ft rows
Lettuce	0.25–0.5 in	8–12 in apart

Note: *Calibrate your planter or drill each season. Check seed metering and depth on a short test run before committing the whole field — a miscalibrated planter can cost yield across every acre.*

4. Water & Irrigation Management

Water is usually the single most limiting factor in crop production. The goal is to keep moisture in the root zone within the range the crop needs — avoiding both drought stress and waterlogging, which suffocates roots and promotes root rots.

Knowing when to water

- **Feel method.** Squeeze soil from root depth; dry, crumbly soil that will not hold shape signals a need for water.
- **Soil moisture sensors.** Tensiometers and capacitance probes give objective readings and can automate irrigation.
- **Evapotranspiration (ET) data.** Local weather networks publish daily ET so you can replace what the crop used.
- **Critical growth stages.** Water stress hurts most during flowering and grain/fruit fill — prioritize water then.

Irrigation methods

Method	Efficiency	Notes
Drip / micro	90–95%	Delivers water to roots; best for row crops, orchards, vegetables; reduces disease by keeping foliage dry.
Sprinkler / center pivot	70–85%	Flexible and widespread for field crops; some loss to wind and evaporation.
Furrow / flood	40–60%	Low cost but inefficient; suited to level fields and certain crops.

Deep, less frequent irrigation encourages deeper rooting and drought resilience compared with frequent shallow watering. Early morning is the best time to irrigate overhead, reducing evaporation and leaf-wetness disease risk.

5. Crop Rotation & Cover Crops

Why rotate

Growing the same crop in the same field year after year builds up crop-specific pests, diseases, and weeds and depletes particular nutrients. Rotation breaks these cycles, spreads nutrient demand, and improves soil structure.

- Alternate crop families so pests and diseases lose their host (e.g., do not follow tomatoes with potatoes — both are nightshades).
- Follow a heavy feeder (corn) with a legume (soybean, clover) that fixes nitrogen and restores fertility.
- Alternate deep- and shallow-rooted crops to use different soil layers and relieve compaction.
- A classic Midwest rotation is corn–soybean; longer rotations adding wheat and a forage legume further reduce pest and weed pressure.

Cover crops

Cover crops are grown between cash crops to protect and improve the soil rather than for harvest. They prevent erosion, suppress weeds, scavenge or fix nutrients, feed soil biology, and break pest cycles.

Type	Examples	Primary benefit
Legumes	Crimson clover, hairy vetch, field peas	Fix nitrogen for the next crop
Grasses / cereals	Cereal rye, oats, annual ryegrass	Scavenge nitrogen, build organic matter, suppress weeds
Brassicas	Tillage radish, mustard	Break compaction, scavenge nutrients, biofumigation

Note: *Terminate cover crops at the right time (mowing, rolling, or herbicide) so they do not compete with the cash crop or go to seed and become weeds themselves.*

6. Weed Management

Weeds compete with crops for light, water, and nutrients, harbor pests, and can slash yields. Effective management is integrated — no single tactic works alone, and over-reliance on one herbicide breeds resistance.

- **Prevention.** Use clean seed, clean equipment between fields, and well-composted manure to avoid introducing weed seed.
- **Cultural.** Narrow rows, vigorous competitive varieties, proper fertility, cover crops, and mulches shade out weeds.
- **Mechanical.** Timely cultivation, hoeing, and mowing kill weeds while they are small; a flush of weeds controlled early prevents a seed bank.
- **Chemical.** Pre-emergent and post-emergent herbicides; rotate herbicide modes of action and tank-mix to slow resistance. Always follow the label.
- **Timing matters most.** The critical weed-free period is the early weeks after crop emergence — keeping fields clean then protects most of the yield.

7. Disease Management

Plant diseases are caused mainly by fungi, bacteria, viruses, and nematodes. They need a susceptible host, a pathogen, and favorable environment — the "disease triangle." Remove any one leg and disease is controlled. (Specific pests and diseases by crop are covered in detail in the companion Pests guide.)

- Choose resistant varieties and certified disease-free seed/transplants.
- Rotate crops to starve soil-borne pathogens of their host.
- Improve air flow and keep foliage dry (proper spacing, drip irrigation, morning watering) to reduce fungal and bacterial spread.
- Remove and destroy infected debris; many pathogens overwinter in residue.
- Scout regularly and act early — fungicides are far more effective as protectants than as rescue treatments.

8. In-Season Scouting & Record-Keeping

Walking fields on a regular schedule is the cheapest insurance in farming. Scouting catches nutrient deficiencies, pests, diseases, and water problems while they are still fixable.

- Scout at least weekly during active growth, following a pattern (e.g., a W or zig-zag) so the whole field is represented, not just the edges.
- Check several plants per stop: look at leaves (top and underside), stems, growing points, and roots.
- Record what you find — date, location, growth stage, problem, and action taken. Records reveal patterns year to year and support better decisions.
- Keep inputs logged (varieties, planting dates, fertilizer, pesticides, irrigation, yields) for planning, compliance, and troubleshooting.

9. Harvest & Post-Harvest Care

Harvesting at the right maturity and handling the crop carefully afterward protects the value built over the whole season. Timing depends on the crop and its end use — grain moisture, fruit color and firmness, or forage stage.

General principles

- Harvest grains at the correct moisture to minimize field loss and spoilage; dry to safe storage moisture (e.g., ~13–15% for most grains) to prevent mold.
- Handle fresh produce gently and cool it quickly; most vegetables and fruits lose quality fastest right after harvest in the heat (“field heat”).
- Clean and sanitize bins, equipment, and storage to avoid carrying over pests and disease.
- Store at the right temperature and humidity for the crop; monitor stored grain for temperature spikes and insects.

Note: *Post-harvest losses can be large and are often avoidable. Prompt drying, cooling, and clean storage frequently matter as much to final returns as any in-field decision.*

10. Crop-Specific Care Notes

Corn

Heavy nitrogen feeder; most sensitive to moisture and heat stress at silking. Side-dress nitrogen around V6 (knee-high). Scout for corn rootworm, European corn borer, and foliar diseases such as gray leaf spot.

Soybean

Legume that fixes much of its own nitrogen with proper inoculation; needs adequate phosphorus, potassium, and sulfur. Watch for soybean cyst nematode, aphids, and white mold. Maintain weed control

early.

Wheat (winter & spring)

Winter wheat is sown in fall, vernalizes over winter, and is harvested early summer; spring wheat is sown as soon as soil can be worked. Top-dress nitrogen in early spring for winter wheat. Scout for rusts, Fusarium head blight, and aphids.

Vegetables (general)

Most benefit from rich, well-drained soil, consistent moisture, and mulch. Transplants (tomato, pepper, brassicas) establish faster than direct seed for long-season crops. Stagger plantings (succession) for a continuous harvest and rotate families each year to limit disease.

Cotton

Warm-season crop needing a long frost-free period and heat. Manage carefully for bollworm/budworm and plant bugs; manage plant growth with regulators to balance vegetative and reproductive growth. Defoliate before mechanical harvest.